08

Interconnecting Networks

Connecting on-premises sites, other clouds and other VPCs — and choosing the right path for each

PCA study series · 9 concept slides · 8 scenario questions

Five ways to link your networks with Google

Option	Path	Bandwidth	Reaches	Encrypted?
HA VPN	IPsec over the internet	~1.5–3 Gbps per tunnel	VPC internal IPs	Yes (IPsec)
Dedicated Interconnect	Your cross-connect in a colocation facility	10 or 100 Gbps links	VPC internal IPs	No by default; add MACsec or HA VPN
Partner Interconnect	Through a service provider	50 Mbps – 50 Gbps	VPC internal IPs	No by default
Cross-Cloud Interconnect	Google-managed link to another cloud	10 or 100 Gbps	VPC ↔ other cloud	Optional MACsec
Direct / Carrier Peering	Exchange with Google's edge	Varies	Google public IPs only	App-level only

EXAM LENS First question to ask: does the traffic need private VPC addresses, or only Google's public services?

Classic VPN versus HA VPN

Classic VPN

Legacy option

- One gateway interface with one external IP
- 99.9% service availability
- Static routing (dynamic routing is deprecated)
- Fine for simple, low-stakes links

HA VPN

Recommended

- Two interfaces, each with its own external IP
- 99.99% SLA when both interfaces connect to the peer with two tunnels
- Requires BGP through Cloud Router
- Active/active or active/passive tunnel use

EXAM LENS '99.99%' + VPN → HA VPN, two tunnels, BGP. One tunnel only earns 99.9%.

Cloud Router makes routing dynamic

1 What it does

A managed BGP speaker that exchanges routes between your VPC and on-premises or other networks. No traffic flows through it.

2 Why it matters

New subnets on either side are advertised automatically — no manual route edits, no tunnel downtime.

3 Regional vs global routing

VPC dynamic routing mode decides whether learned routes reach only the router's region or every region in the VPC.

4 Where it's required

HA VPN, Dedicated and Partner Interconnect, and Cross-Cloud Interconnect all rely on it.

EXAM LENS 'On-premises can't reach a subnet in another region' → switch the VPC to global dynamic routing.

Dedicated or Partner Interconnect

Dedicated Interconnect

You meet Google in a colocation facility

- 10 Gbps or 100 Gbps circuits, bundled for more
- Your router must sit in a supported colocation facility
- 99.9% or 99.99% SLA depending on topology
- Best value for sustained, very high bandwidth

Partner Interconnect

A provider bridges the distance

- Capacities from 50 Mbps to 50 Gbps
- No need to reach a Google colocation facility
- Layer 2 or Layer 3 options from the partner
- Best when sites are remote or bandwidth needs are moderate

EXAM LENS Neither Interconnect encrypts by default. 'Must be encrypted' → MACsec or HA VPN over Cloud Interconnect.

Reaching other clouds and Google's public services

Not every link needs to land in a VPC. Pick the path by destination.

Cross-Cloud Interconnect	Dedicated, Google-provisioned links to AWS, Azure, Oracle Cloud or Alibaba Cloud.
Direct Peering	Exchange traffic with Google at a public peering point to reach Workspace and Google APIs.
Carrier Peering	The same reach via a service provider when you can't peer directly.
Peering limits	No SLA, and no access to VPC internal IPs.

Choosing a connection: requirement first, product second

Requirement	Choose
Private VPC access, quick to set up, encrypted, < a few Gbps	HA VPN
Private VPC access, 10 Gbps+, data centre in a Google colocation facility	Dedicated Interconnect
Private VPC access, remote site or 50 Mbps–50 Gbps	Partner Interconnect
High-bandwidth private link to another cloud	Cross-Cloud Interconnect
Only need Google Workspace or public APIs, no SLA required	Direct or Carrier Peering
Encryption over Interconnect	MACsec, or HA VPN over Cloud Interconnect

Shared VPC: one network, many projects

1 Host project

Owns the VPC, subnets, routes and firewall rules. Administered by a central network team.

2 Service projects

Attach to the host and run their VMs and services on the shared subnets.

3 Granular sharing

Grant Network User on specific subnets, so each team can deploy only where allowed.

4 Boundary

Host and service projects must belong to the same organisation.

EXAM LENS 'Centralised network control, decentralised app teams' → Shared VPC.

VPC Network Peering: two networks, private connectivity

1 Across boundaries

Connects VPCs in the same or different projects and organisations using internal IPs.

2 Decentralised

Each side keeps its own admins, firewall rules and routes.

3 Not transitive

If A peers with B and B with C, A still cannot reach C. Peer directly or use a hub such as Network Connectivity Center.

4 No overlap

Subnet ranges must not overlap — plan IP space before mergers or partnerships.

EXAM LENS 'Two companies after an acquisition, different orgs' → VPC Network Peering, not Shared VPC.

Shared VPC and VPC peering compared

	Shared VPC	VPC Network Peering
Across organisations	No	Yes
Within one project	No (host and service projects)	Yes — two VPCs in one project
Network administration	Centralised in the host project	Each VPC administers itself
Transitive	Not applicable — it's one network	No
Typical use	Enterprise landing zone	Partners, acquisitions, SaaS producers



8 PCA-style scenarios

Read every requirement word, eliminate options that break one, then pick the most managed answer that satisfies them all. Each question is followed by its answer and the reasoning.

Question 1 of 8

VPN availability

A company needs an encrypted connection from its data centre to a VPC with a 99.99% availability SLA. Bandwidth needs are under 2 Gbps and the link must be live in days.

A Classic VPN with two tunnels and static routes

B HA VPN with both interfaces connected to two on-premises VPN devices and BGP through Cloud Router

C Partner Interconnect at 1 Gbps

D Direct Peering with application-level TLS

Answer 1: B

B HA VPN with both interfaces connected to two on-premises VPN devices and BGP through Cloud Router

Why it fits

HA VPN is the only VPN option with a 99.99% SLA, which requires both interfaces in use and dynamic routing. It's encrypted and fast to set up.

Why the others don't

A — Classic VPN caps at 99.9%.

C — Not encrypted by default and slower to provision.

D — Does not reach VPC internal IPs.

EXAM LENS '99.99%' + 'VPN' → HA VPN, two tunnels, BGP.

Question 2 of 8

Hybrid bandwidth

Your data centre is in a colocation facility where Google has a point of presence. You need 40 Gbps of private connectivity to VPC resources for nightly replication.

A. Twenty HA VPN tunnels

B. Dedicated Interconnect with four 10 Gbps circuits (or one 100 Gbps)

C. Partner Interconnect at 50 Mbps

D. Carrier Peering

Answer 2: B

B Dedicated Interconnect with four 10 Gbps circuits (or one 100 Gbps)

Why it fits

Being in a supported facility means you can cross-connect directly. Dedicated Interconnect delivers high, consistent bandwidth to internal IPs at the best price for this volume.

Why the others don't

A — Internet-bound, complex and less predictable.

C — Far too little bandwidth.

D — Reaches only Google public IPs.

EXAM LENS 'Colocation facility' + '10 Gbps+' → Dedicated Interconnect.

Question 3 of 8

Remote site

A branch office in a city with no Google colocation facility needs a 2 Gbps private link to a VPC with an SLA, through an existing telecom provider.

A Dedicated Interconnect

B Partner Interconnect

C Direct Peering

D Cloud NAT

Answer 3: B

B Partner Interconnect

Why it fits

A supported service provider extends connectivity to Google, so the branch gets private, SLA-backed bandwidth without reaching a colocation facility.

Why the others don't

A — Requires presence in a Google colocation facility.

C — Public IPs only and no SLA.

D — Provides outbound internet for private VMs; unrelated.

EXAM LENS 'Can't reach colocation' / 'through a provider' → Partner Interconnect.

Question 4 of 8

Encryption

Security policy says all traffic leaving the data centre must be encrypted. You already use Dedicated Interconnect for its bandwidth. What should you do?

A. Nothing — Interconnect traffic is encrypted by default

B. Enable MACsec for Cloud Interconnect, or run HA VPN over Cloud Interconnect

C. Switch to Direct Peering

D. Put a Cloud NAT gateway in front of the VPC

Answer 4: B

B Enable MACsec for Cloud Interconnect, or run HA VPN over Cloud Interconnect

Why it fits

Interconnect provides a private path but not encryption. MACsec encrypts at Layer 2 on the link; HA VPN over Interconnect adds IPsec while keeping the private circuit.

Why the others don't

A — Incorrect: the default is unencrypted.

C — Different reach; doesn't address the policy.

D — NAT handles address translation, not encryption.

EXAM LENS Private ≠ encrypted. Interconnect + encryption → MACsec or HA VPN over Interconnect.

Question 5 of 8

Shared VPC

A central network team must own all subnets and firewall rules. Twenty application teams, each with their own projects in the same organisation, must deploy VMs into approved subnets only.

A VPC Network Peering between twenty VPCs

B Shared VPC with a host project and service projects, granting Network User on specific subnets

C One project for all teams with Editor for everyone

D HA VPN between each team's VPC

Answer 5: B

B Shared VPC with a host project and service projects, granting Network User on specific subnets

Why it fits

Shared VPC centralises network administration in the host project while subnet-level Network User grants let each team deploy only where permitted.

Why the others don't

A — Decentralises control and doesn't scale; each VPC has its own admins.

C — Removes project isolation and violates least privilege.

D — Unnecessary complexity and cost inside one organisation.

EXAM LENS 'Same organisation' + 'central network control' → Shared VPC.

Question 6 of 8

Cross-org

After an acquisition, your VPC and the acquired company's VPC — in a different organisation, with non-overlapping ranges — must communicate privately with low latency.

A Shared VPC

B VPC Network Peering

C Direct Peering

D External passthrough load balancer

Answer 6: B

B VPC Network Peering

Why it fits

Peering connects VPCs across organisations using internal IPs with Google-network latency, and each company keeps its own administration.

Why the others don't

A — Requires the same organisation.

C — Connects to Google's edge, not to another VPC.

D — Exposes services publicly.

EXAM LENS 'Different organisations' → VPC peering.

Question 7 of 8

Transitivity

VPC-A peers with VPC-B, and VPC-B peers with VPC-C. VMs in VPC-A cannot reach VMs in VPC-C, even though firewall rules allow it. Why?

A. Firewall rules must be created in VPC-B

B. VPC Network Peering is not transitive; A and C need their own connection

C. Peering only supports TCP port 80

D. Cloud Router in VPC-B is missing

Answer 7: B

B VPC Network Peering is not transitive; A and C need their own connection

Why it fits

Routes learned via peering aren't re-exported to other peers. A must peer with C directly or use a hub like Network Connectivity Center.

Why the others don't

A — B's firewall doesn't route traffic between other peers.

C — Peering supports all protocols.

D — Peering doesn't use Cloud Router.

EXAM LENS Peering never chains. Hub-and-spoke needs NCC or a different design.

Question 8 of 8

Multicloud

An analytics workload in Google Cloud must pull large volumes of data nightly from a database running in AWS over a private, high-bandwidth link without managing colocation equipment.

A HA VPN over the internet

B Cross-Cloud Interconnect

C Carrier Peering

D Partner Interconnect from AWS

Answer 8: B

B Cross-Cloud Interconnect

Why it fits

Cross-Cloud Interconnect provides Google-provisioned, dedicated 10 or 100 Gbps links directly between Google Cloud and other cloud providers.

Why the others don't

A — Lower bandwidth over the public internet.

C — Reaches Google public services only.

D — Designed for on-premises, not another cloud.

EXAM LENS 'Another cloud' + 'high bandwidth' → Cross-Cloud Interconnect.